46. "If compound X is boiling, then its temperature must be at least 150°C." Assuming that this statement is true, which of the following must also be true?

a. If the temperature of compound X is at least 150°C, then compound X is boiling.

b. If the temperature of compound X is less than 150°C, then compound X is not boiling.

c. Compound X will boil only if its temperature is at least 150°C.

d. If compound X is not boiling, then its temperature is less than 150°C.

e. A necessary condition for compound X to boil is that its temperature be at least 150°C.

f. A sufficient condition for compound X to boil is that its temperature be at least 150°C.

Let's start by assigning variables:

p = Compond X is boiling
q = The temperature is at least 150°C

Then we can translate the conditional statement to symbolic form:

Statement	**Symbolic form**
"If compound X is boiling, then its temperature must be at least 150°C."	$p \rightarrow q$

And now we can translate the individual statements from a through f.

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	**Symbolic form**
a. If the temperature of compound X is at least 150°C, then compound X is boiling.	$q \rightarrow p$

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	**Symbolic form**
b. If the temperature of compound X is less than 150°C, then compound X is not boiling.	$\sim q \to \sim p$

Let's look at the definition of the contrapositive of a conditional statement (according to Epp):

Definition

The **contrapositive** of a conditional statement of the form "If p then q" is

If ~q then ~p.

Symbolically,

The contrapositive of p S → q is ~q → ~p.

By this definition, $\sim q \to \sim p$ should be logically equivalent to $p \to q$. We can prove this using a truth table:

p	**q**	**p → q**	**~p**	**~q**	**~q → ~p**
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

We can now conclude that $\sim q \to \sim p \equiv p \to q$

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	**Symbolic form**
c. Compound X will boil only if its temperature is at least 150°C.	$p \rightarrow q$

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	**Symbolic form**
d. If compound X is not boiling, then its temperature is less than 150°C.	$\sim p \rightarrow \sim q$

We can apply the same definition of the contrapositive here for $\sim p \rightarrow \sim q$, which’s $q \rightarrow p$

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	Symbolic form
e. A necessary condition for compound X to boil is that its temperature be at least 150°C.	$p \rightarrow q$

p = Compond X is boiling
q = The temperature is at least 150°C

Statement	Symbolic form
f. A sufficient condition for compound X to boil is that its temperature be at least 150°C.	$q \rightarrow p$

Now that we have all statements and their corresponding symbol form, we can list them together for reference:

Assumed to be true	Symbolic form
If compound X is boiling, then its temperature must be at least 150°C.	$p \to q$

Statement	Symbolic form
a. If the temperature of compound X is at least 150°C, then compound X is boiling.	$q \to p$
b. If the temperature of compound X is less than 150°C, then compound X is not boiling.	$\sim q \to \sim p \equiv p \to q$
c. Compound X will boil only if its temperature is at least 150°C.	$p \to q$
d. If compound X is not boiling, then its temperature is less than 150°C.	$\sim p \to \sim q$ is $q \to p$
e. A necessary condition for compound X to boil is that its temperature be at least 150°C.	$p \to q$
f. A sufficient condition for compound X to boil is that its temperature be at least 150°C.	$q \to p$

Now that we have the list of statements in their symbolic form, we can also assess that **a**, **d** and **f** are logically equivalent to each other, whereas **b**, **c** and **e** are logically equivalent to each other.

Let's compare the statement forms in a truth table to see whether they are logically equivalent in turn:

p	**q**	**p → q**	**q → p**
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

By examining the truth sets for each statement form, we can now conclude that they are not logically equivalent.

And since the initial statement is assumed to the true, then that also means **b**, **c** and **e** are also assumed to be true, since they are logically equivalent to each other (see previous page index).

We conclude to assume then that **a**, **d** and **f** does not have to be true.

Solution: **b**, **c** and **e** are assumed to be true, whereas **a**, **d** and **f** are assumed not to be true.